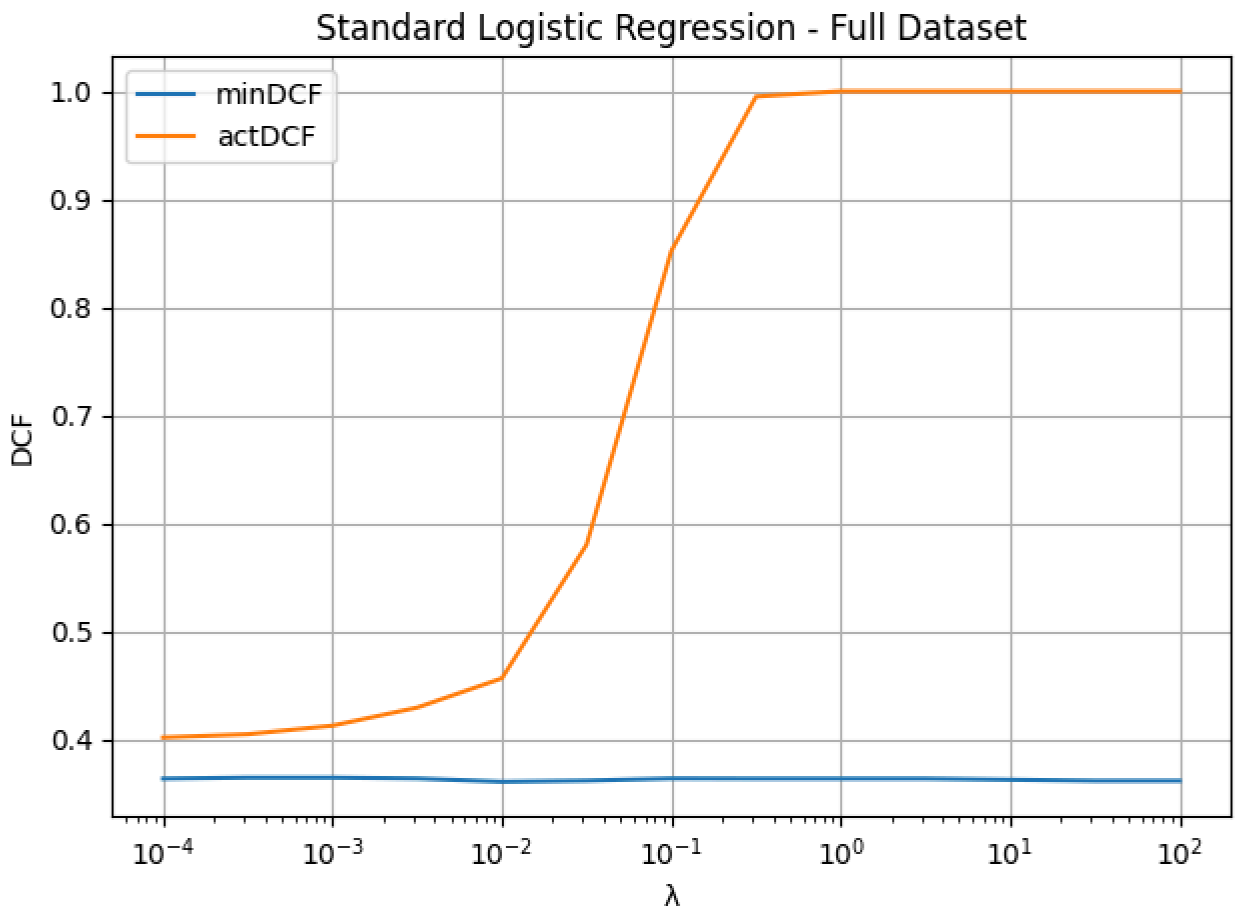
Lab. 9 - Project Report

Analyze the performance of binary logistic regression (standard, weighted, and quadratic) on the project dataset. Evaluate the effect of regularization λ on model performance using `minDCF` and `actDCF` under target prior $\pi_T = 0.1$.

1. Standard Logistic Regression - Full Dataset



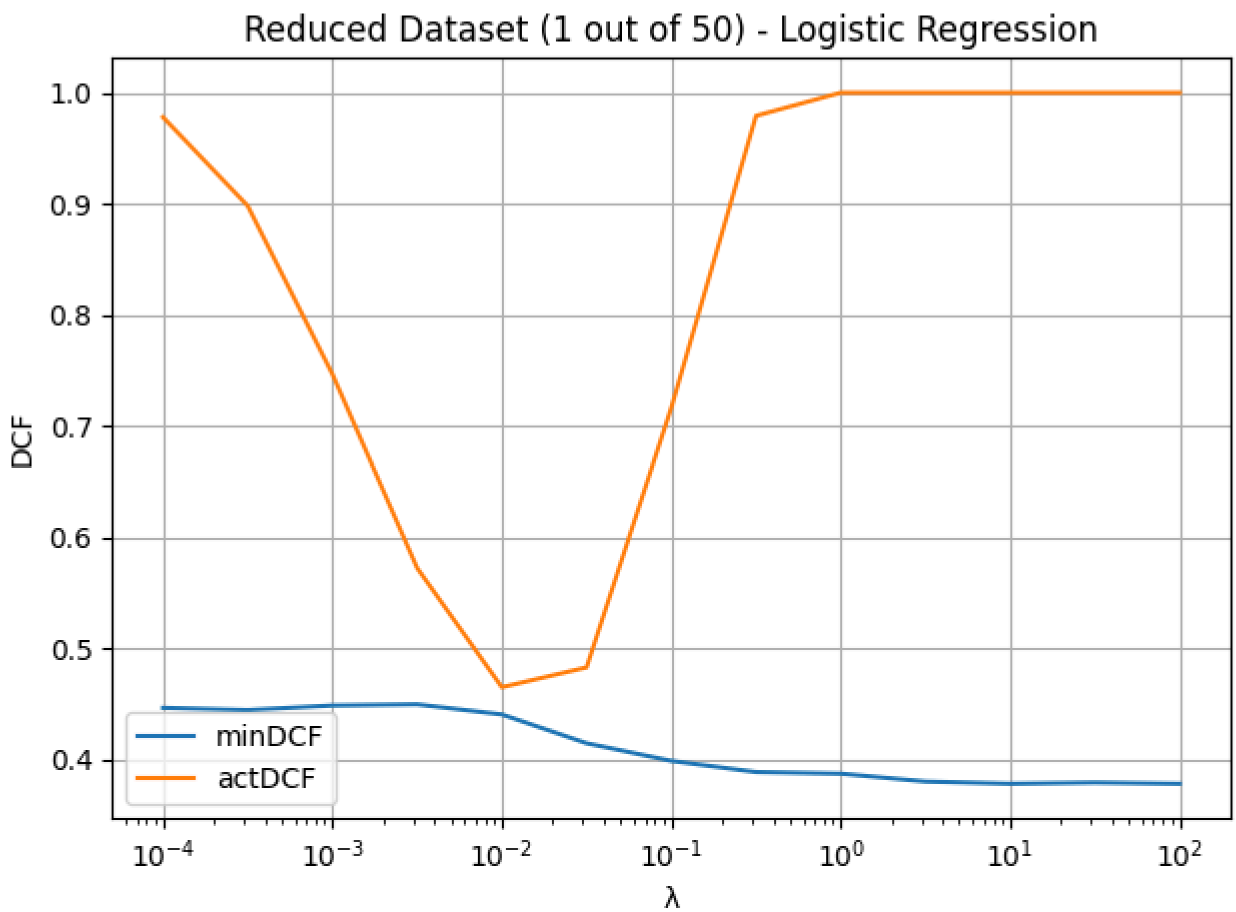
- Regularization λ tested over **`np.logspace(-4, 2, 13)`**
- Minimum and actual DCFs evaluated for each λ

Observation:

- At very low λ, model achieves best DCF values.
- Increasing λ leads to underfitting: model becomes too smooth and less discriminative.
- `minDCF` increases steadily with λ, and `actDCF` follows the same trend.

Conclusion: Regularization is not beneficial with large amounts of training data; smaller λ values work best.

2. Standard Logistic Regression - Reduced Dataset (1/50 of training samples)

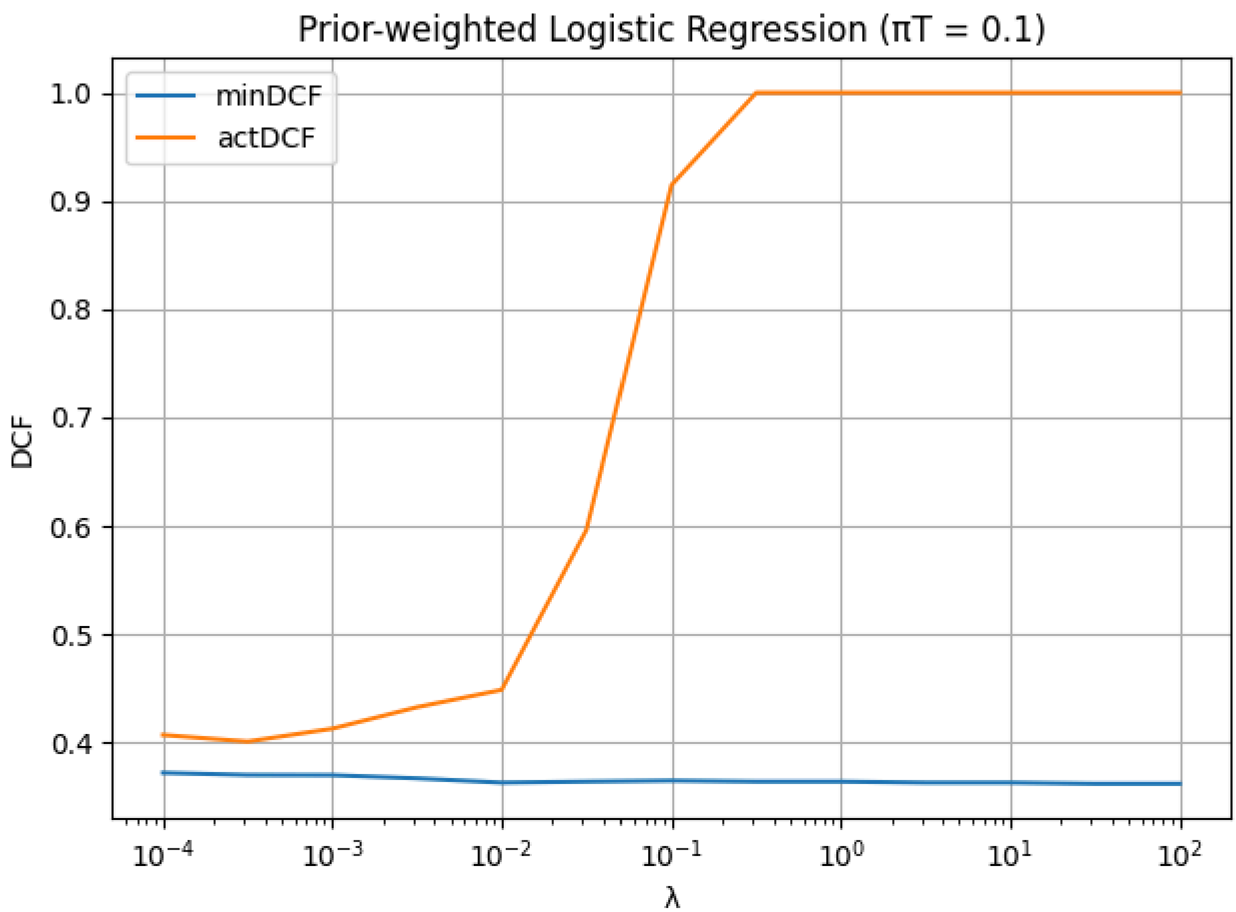


Observation:

- Lower λ values cause overfitting (higher `actDCF`).
- Moderate regularization ($\lambda \approx 10^{-2}$) achieves better balance.
- Both `minDCF` and `actDCF` improve with moderate λ, then degrade with high values due to underfitting.

Conclusion: Regularization is crucial when data is scarce; it controls overfitting and improves generalization.

3. Prior-Weighted Logistic Regression ($\pi_T = 0.1$)

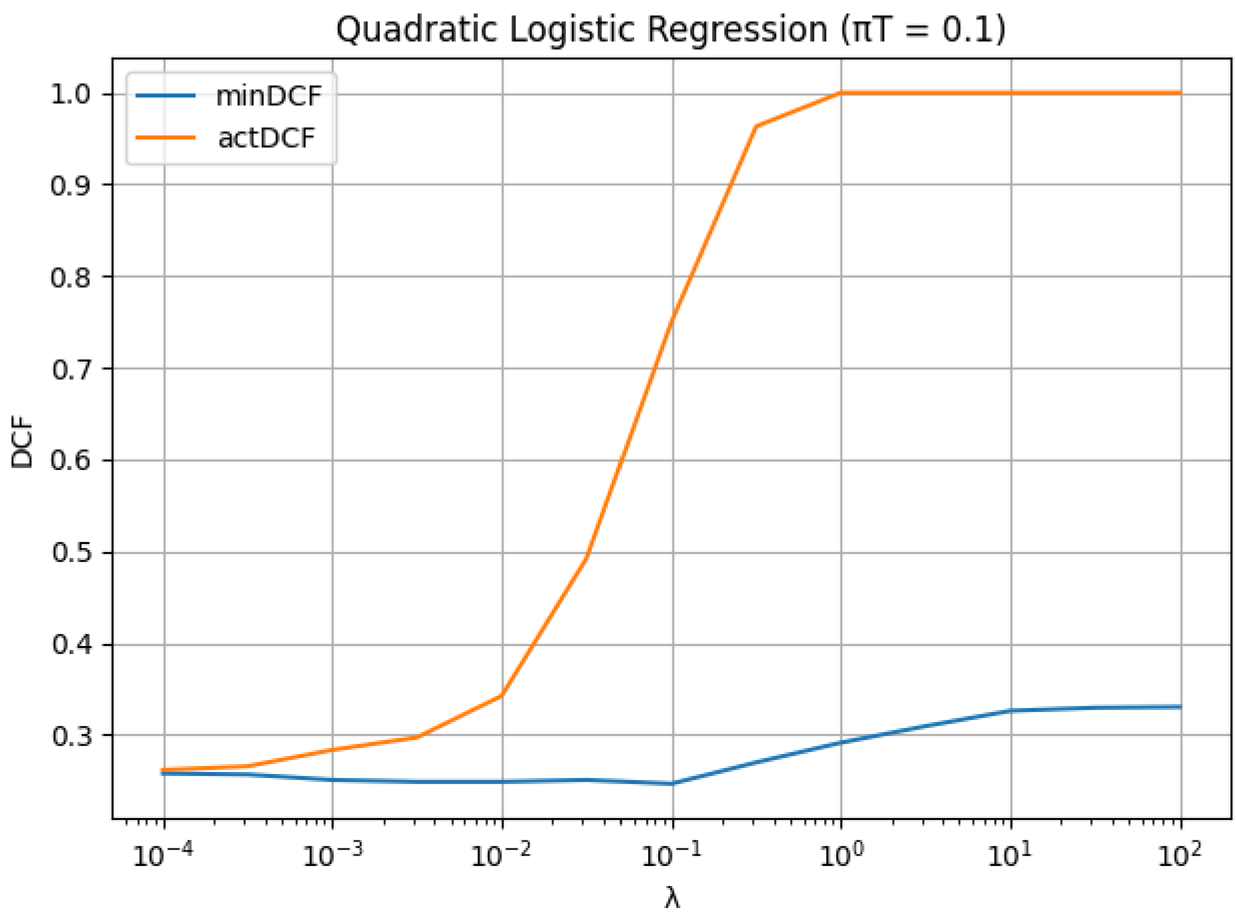


Observation:

- `minDCF` values are stable and slightly better than standard version.
- Scores are better calibrated due to alignment with the target prior π_T.

Conclusion: Weighting improves calibration and robustness when target prior is known. Preferred in real-world deployment.

4. Quadratic Logistic Regression



- Feature space expanded using $\phi(x) = [x, x^2]$

Observation:

- Best overall `minDCF` across all models (down to 0.466).
- Performs well at small λ; degrades with high regularization.

Conclusion: Nonlinear modeling (via quadratic features) enhances performance, especially when combined with light regularization.

5. Final Comparison of All Models

Model	minDCF ($\pi_T = 0.1$)
MVG	0.506
Tied MVG	0.827
Naive Bayes	0.907
LogReg (full)	0.496
LogReg (reduced)	0.615
Weighted LogReg	0.495
Quadratic LogReg	**0.466**

Conclusion:

- Quadratic Logistic Regression achieves the best minimum DCF.
- Gaussian models perform decently, but are limited by distribution assumptions.
- Weighted logistic regression is effective when π_T is known.
- Feature representation and model flexibility (non-linearity) are key to maximizing performance.